



MAHARAJA AGRASEN UNIVERSITY

Mid Semester Test-II

Semester II, May 2025

Course: BCA

Subject: Mathematics-II

Subject Code: BCA-202

Time: 1.5 Hours

Max. Marks: 20

- Instructions:- i) Attempt **THREE** questions in all.
 ii) **Part A is compulsory.** Attempt any **FIVE** parts from it.
 iii) Attempt **TWO** questions from Part B by selecting one question from each section.

PART A

Q1. (i) Define a continuous function.

(ii) Evaluate Left hand and Right hand limit of the function $f(x) = 4x + 5$ at $x = 0$.

(iii) State Intermediate mean value theorem.

(iv) Define three types of discontinuities for a function $f(x)$ at $x = a$.

(v) Evaluate $\frac{d}{dx}(x^2 + 1)$ and $\frac{d}{dx}(\cot x)$.

(vi) Evaluate derivative of $\sin(10x^2)$ with respect to x .

(vii) Evaluate derivative of $e^{(4x+5)}$ with respect to x .

PART B Section I

Q2. Discuss the continuity of the functions

(a) $f(x) = \begin{cases} \frac{|x-2|}{x-2}, & \text{when } x \neq 2 \\ 1, & \text{when } x = 2 \end{cases}$ at $x = 2$.

(b) $f(x) = \begin{cases} 1+x, & \text{when } x \leq 2 \\ 5-x, & \text{when } x > 2 \end{cases}$ at $x = 2$.

(5)

Q3. State the type of discontinuity of the functions

(a) $f(x) = \begin{cases} \frac{\sin 5x}{x}, & \text{when } x \neq 0 \\ 1, & \text{when } x = 0 \end{cases}$ at $x = 0$.

(b) $f(x) = \begin{cases} \frac{|x|}{x}, & \text{when } x \neq 0 \\ 1, & \text{when } x = 0 \end{cases}$ at $x = 0$.

(5)

II kind

Section II

Q4. Evaluate the derivative of the following functions with respect to x

(a) $(x^2 + 6x + 8)^7$

(b) $\sin^2 x$

(c) xe^x

(d) $x \log x$

(5)

Q5. Find the maximum and minimum values of the function $2x^3 - 3x^2 - 36x + 10$

(5)

$$\frac{72}{12} = 6$$

$$\frac{27}{18} = \frac{3}{2}$$

$$\frac{108}{11}$$

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